Part 1 — Setup: MetaMask Sepolia + Test ETH

Task 1.1: Enable Sepolia in MetaMask

1. Open MetaMask.
2. Go to **Settings** → **Advanced**.
3. Turn ON **Show test networks**.
4. Open the network dropdown and select **Sepolia**.

Checkpoint: MetaMask network shows **Sepolia**.

Task 1.2: Get Sepolia ETH from a Faucet

Use any faucet you prefer. Two common options:

- MetaMask testnet faucet
- Chainlink Sepolia faucet

1. Copy your MetaMask wallet address.
2. Request test ETH from the faucet.
3. Wait until the balance updates.

Checkpoint: You have at least **0.2 Sepolia ETH** (enough for deploy + multiple mint tests).

Part 2 — Remix Setup: Create the Contract File

Task 2.1: Open Remix and Create a File

1. Go to **remix.ethereum.org**
2. In **File Explorer**, click **New File**
3. Name it:
   SimpleMintContract.sol
4. Paste the contract code (exactly as given).

Checkpoint: File saved and visible in the explorer.

Part 3 — Dependencies: OpenZeppelin Imports

Task 3.1: Resolve OpenZeppelin Imports

Your contract uses:

- Ownable.sol
- ERC721URIStorage.sol

1. Click **Solidity Compiler** tab.
2. Click **Compile** once.
3. If you see import errors (OpenZeppelin not found):
    - Go to **File Explorer**
    - Look for **Dependencies / Remix Packages**
    - Add/install **OpenZeppelin Contracts**

Checkpoint: The contract compiles without import errors.

Part 4 — Compile the Contract

Task 4.1: Compile Settings

1. In **Solidity Compiler**:
    - Set compiler version to a **0.8.x** version (example: 0.8.20+)
    - Ensure it matches pragma solidity ^0.8.0;
2. Click: **Compile SimpleMintContract.sol**

Checkpoint: "Compilation successful" appears and no red errors show.

Part 5 — Connect Remix to MetaMask (Injected Provider)

Task 5.1: Use Injected Provider

1. Go to **Deploy & Run Transactions**
2. Under **Environment**, select:
    - **Injected Provider – MetaMask** (or "Injected Web3" depending on UI)

3. MetaMask will pop up:
    - Approve connection
4. Confirm MetaMask is still on **Sepolia**

Checkpoint: Remix shows your MetaMask address under “Account”.

Part 6 — Deploy the ERC-721 Contract

Task 6.1: Deploy

1. In Remix **Deploy** section:
    - Contract: SimpleMintContract
2. Click **Deploy**
3. MetaMask pops up → Confirm

Checkpoint: Contract appears under **Deployed Contracts** in Remix.

Task 6.2: Validate Initial Contract State (Read Calls)

In **Deployed Contracts**, click these **read** functions:

Function	Expected Value
owner()	your MetaMask address
isMintEnabled()	false
maxSupply()	2
totalSupply()	0
mintAmount()	50000000000000000 wei (0.05 ETH)

Checkpoint: All values match.

Part 7 — Functional Test Cases (Minting Rules)

Test Case 1 — Mint while Disabled (Should Fail)

Task 7.1: Attempt Mint (Expect Revert)

1. Expand mint()
2. In Remix top bar, set:
    - **Value:** 0.05
    - **Unit:** ether
3. Click **mint()**
4. Confirm transaction if MetaMask asks (it may fail before prompting depending on UI)

Expected Result: Transaction fails with:

- **"Minting not enabled"**

Record: Write the revert message in your lab notes.

Test Case 2 — Enable Minting (Owner Only)

Task 7.2: Toggle Minting

1. Click toggleisMintEnabled()
2. Confirm in MetaMask
3. Call isMintEnabled() again

Expected Result:

- isMintEnabled() → true

Test Case 3 — Successful Mint (Correct Payment)

Task 7.3: Mint First NFT

1. Set **Value = 0.05 ether**
2. Click mint()
3. Confirm MetaMask

Expected Result (after mined):

- totalSupply() → 1

- mintedWallets(yourAddress) → 1

Record: Transaction hash + updated values.

Test Case 4 — Mint Again Same Wallet (Should Fail)

Task 7.4: Attempt 2nd Mint from Same Wallet

1. Set **Value = 0.05 ether**
2. Click mint()

Expected Result: Revert message: **"Exceeds max per wallet"**

Test Case 5 — Max Supply Behavior (Need 2nd Wallet)

Because:

- maxSupply = 2
- each wallet can only mint **1**

Task 7.5: Mint Using Another MetaMask Account

1. In MetaMask, switch to **Account 2**
2. Ensure Account 2 has Sepolia ETH
3. Go back to Remix (Injected Provider automatically uses active account)
4. Call mint() with **0.05 ether**

Expected Result: totalSupply() → 2

Test Case 6 — Sold Out (Should Fail)

Task 7.6: Attempt Another Mint (Any Wallet)

Try minting again with any account.

Expected Result: Revert message: **"Sold out"**

Part 8 — Common Gotchas (Troubleshooting Checklist)

Issue A: Wrong Network

Symptom: Transactions fail / contract not visible / wrong chain

- Fix: Ensure MetaMask network is **Sepolia**
- Reminder: Injected Provider follows MetaMask network

Issue B: Wrong Value Sent

Symptom: Revert "Wrong value"

- Fix: Your contract requires:
    - msg.value == mintAmount exactly
- Use **0.05 ether** or **50000000000000000 wei**

Issue C: ERC721URIStorage "Unused"

Observation: You imported URI storage but did not set token URIs.

- Result: You can mint NFTs, but **they won't have metadata URIs** unless you implement _setTokenURI.

Part 9 — Post-Lab Questions (Assessment)

Answer briefly:

1. Why did mint() fail before enabling minting?

   Because isMintEnabled was false by default. The require statement checks this value, and since minting wasn't enabled by the owner, the transaction reverted with "Minting not enabled."

2. What specific line enforces "1 mint per wallet"?

   The line require(mintedWallets[msg.sender] < 1, "Exceeds max per wallet"); enforces the limit. It prevents a wallet from minting more than once.

3. What happens if you send 0.051 ETH?

   he transaction reverts with "Wrong value" because the contract requires the payment to be exactly 0.05 ETH using strict equality (==).

4. Why do you need a second wallet to reach maxSupply = 2?

   Each wallet can mint only one NFT. Since maxSupply is 2, a second wallet is required to mint the second NFT.

5. What is the value of mintAmount in **wei**?

   50,000,000,000,000,000 wei

Part 10 — Optional Extension (Metadata Upgrade Challenge)

Challenge Goal

Upgrade the contract so each minted token has a metadata URI.

Suggested approach

- Modify mint() to accept a URI:
  - mint(string memory uri) external payable
- After _safeMint, call:
  - _setTokenURI(tokenId, uri);

Success criteria:

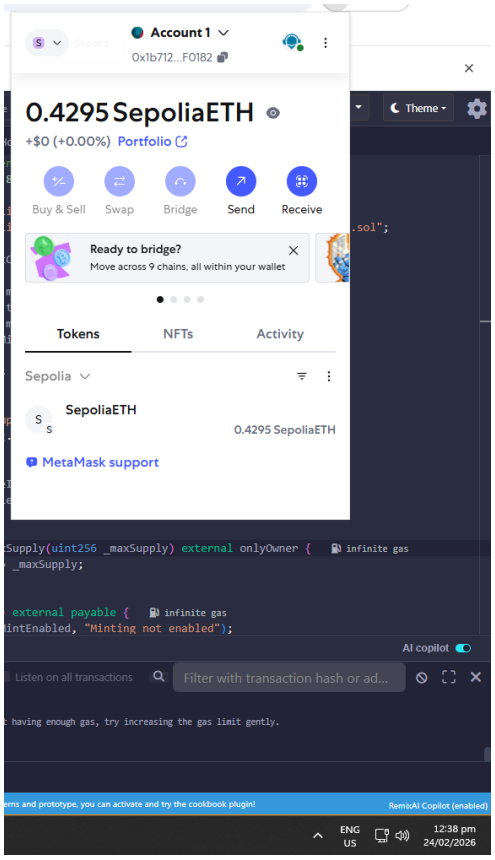
- After minting, calling tokenURI(tokenId) returns the URI you provided.

Student Output Checklist (What to Submit)

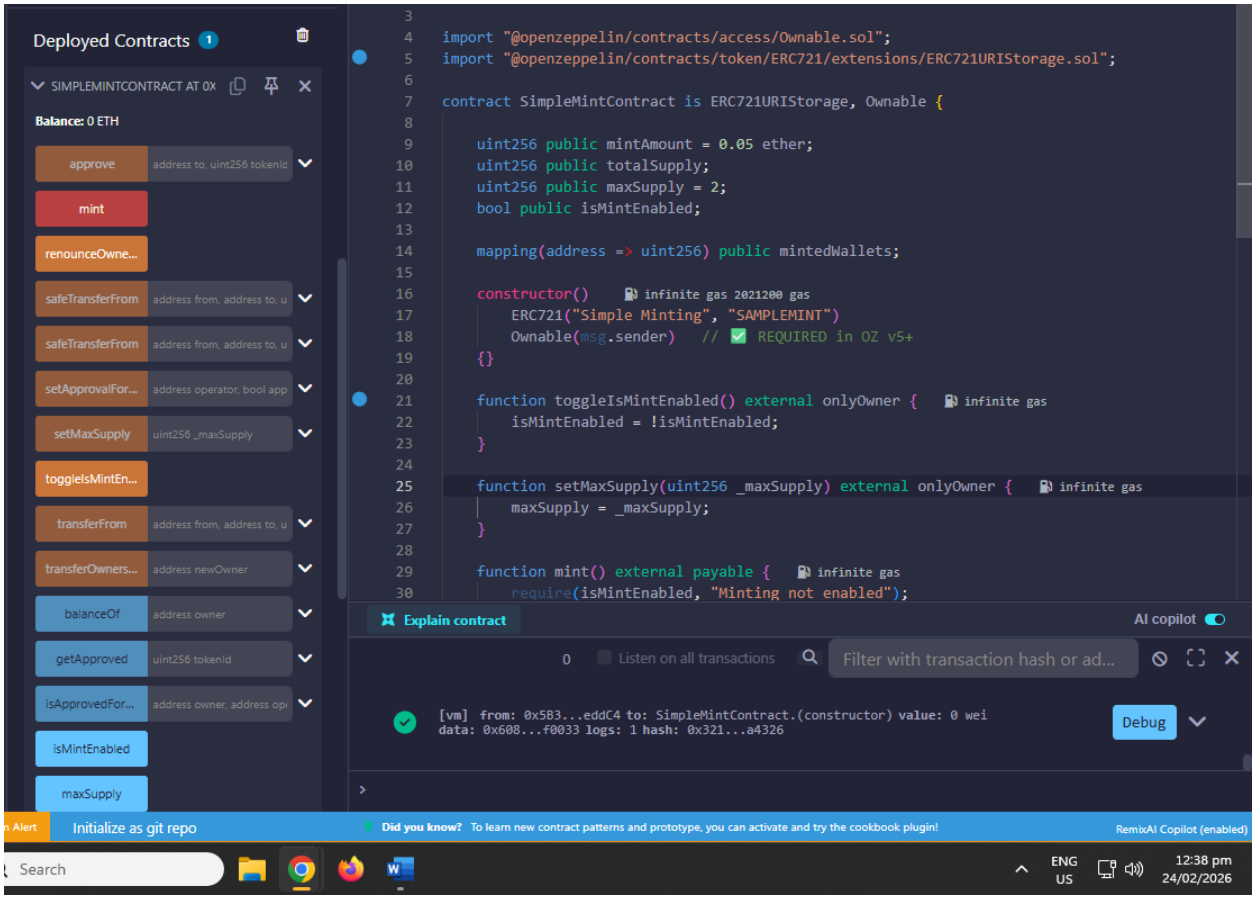
Screenshots of the following:

1.

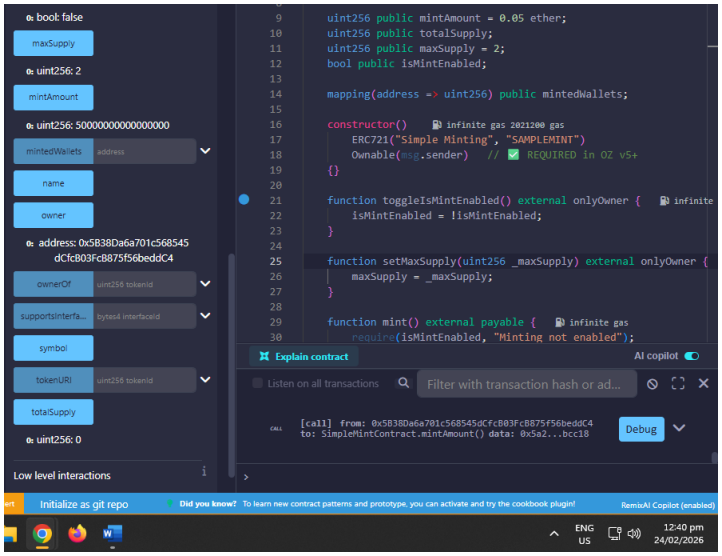
    1. MetaMask showing Sepolia network + balance



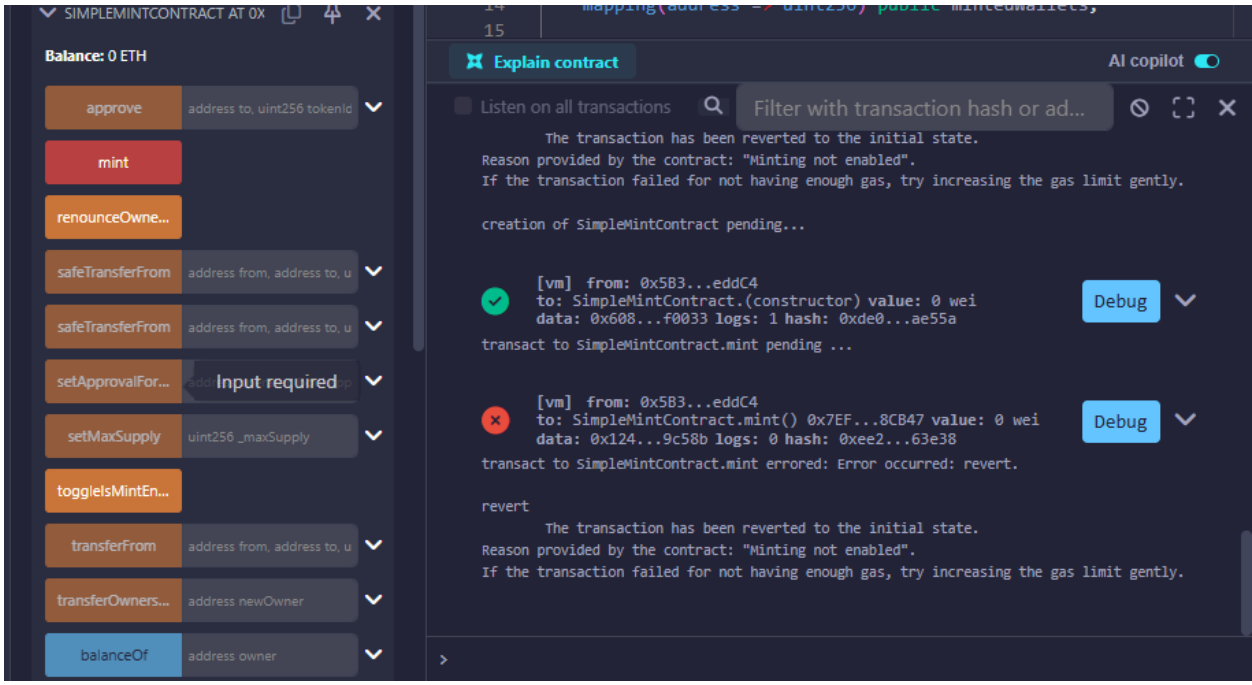
    2. Remix deployed contract

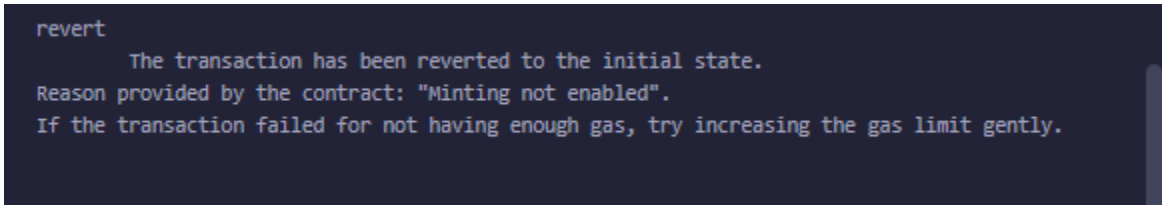


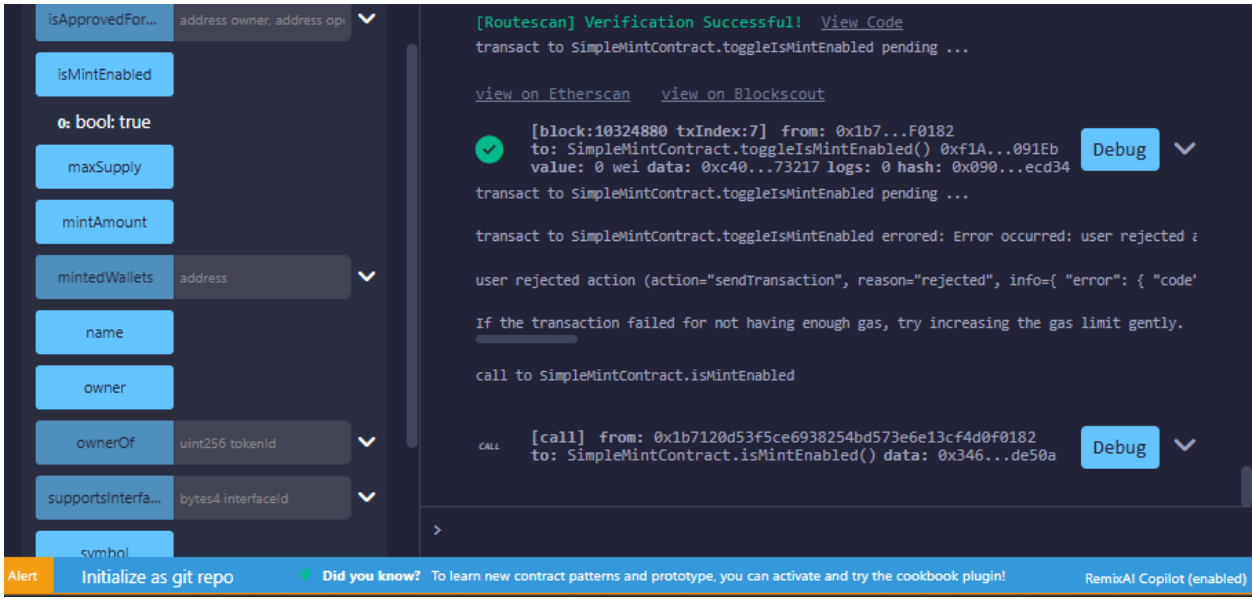
3. Successful mint transaction (1st wallet)



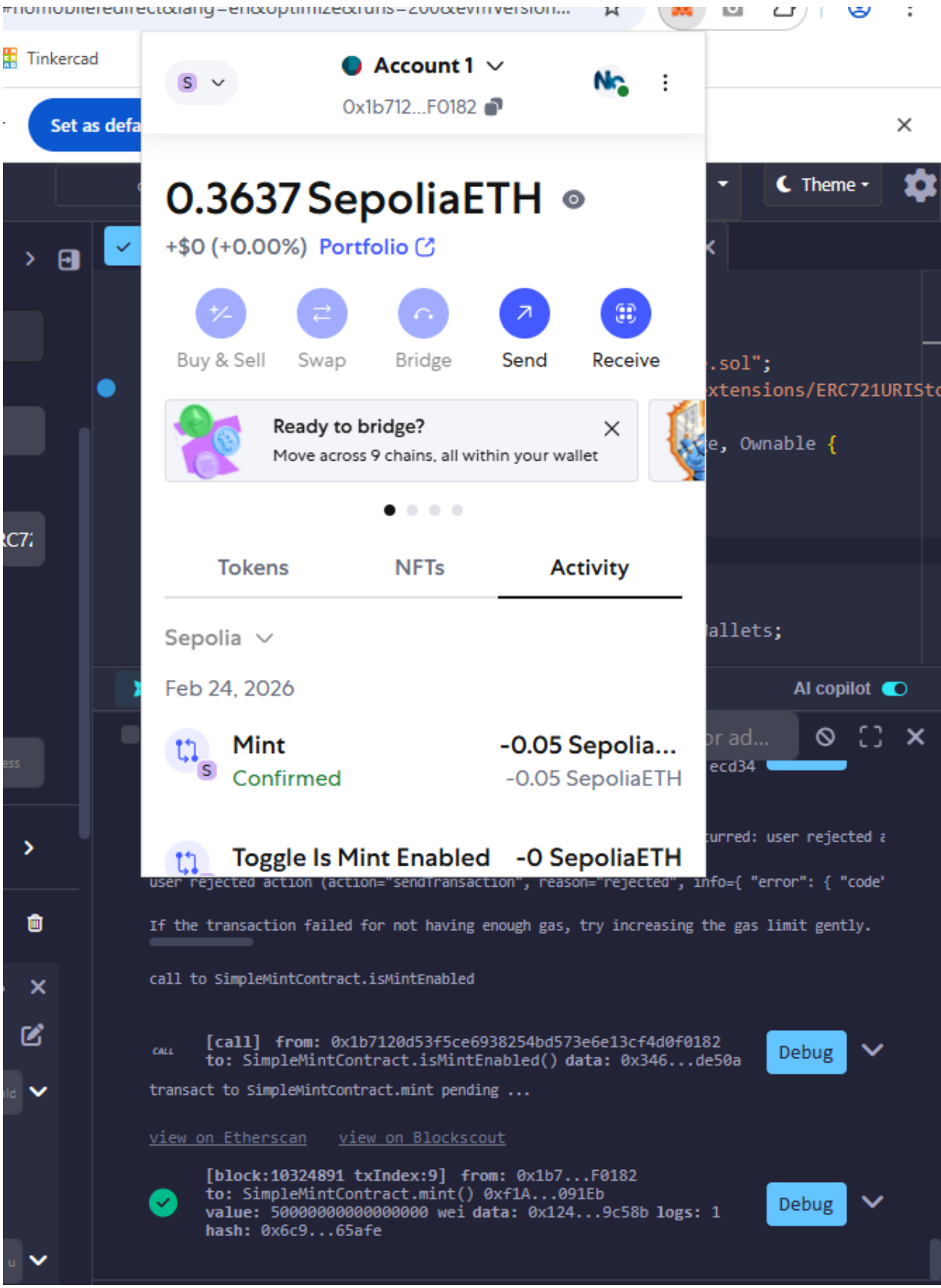
4. Revert: “Minting not enabled”



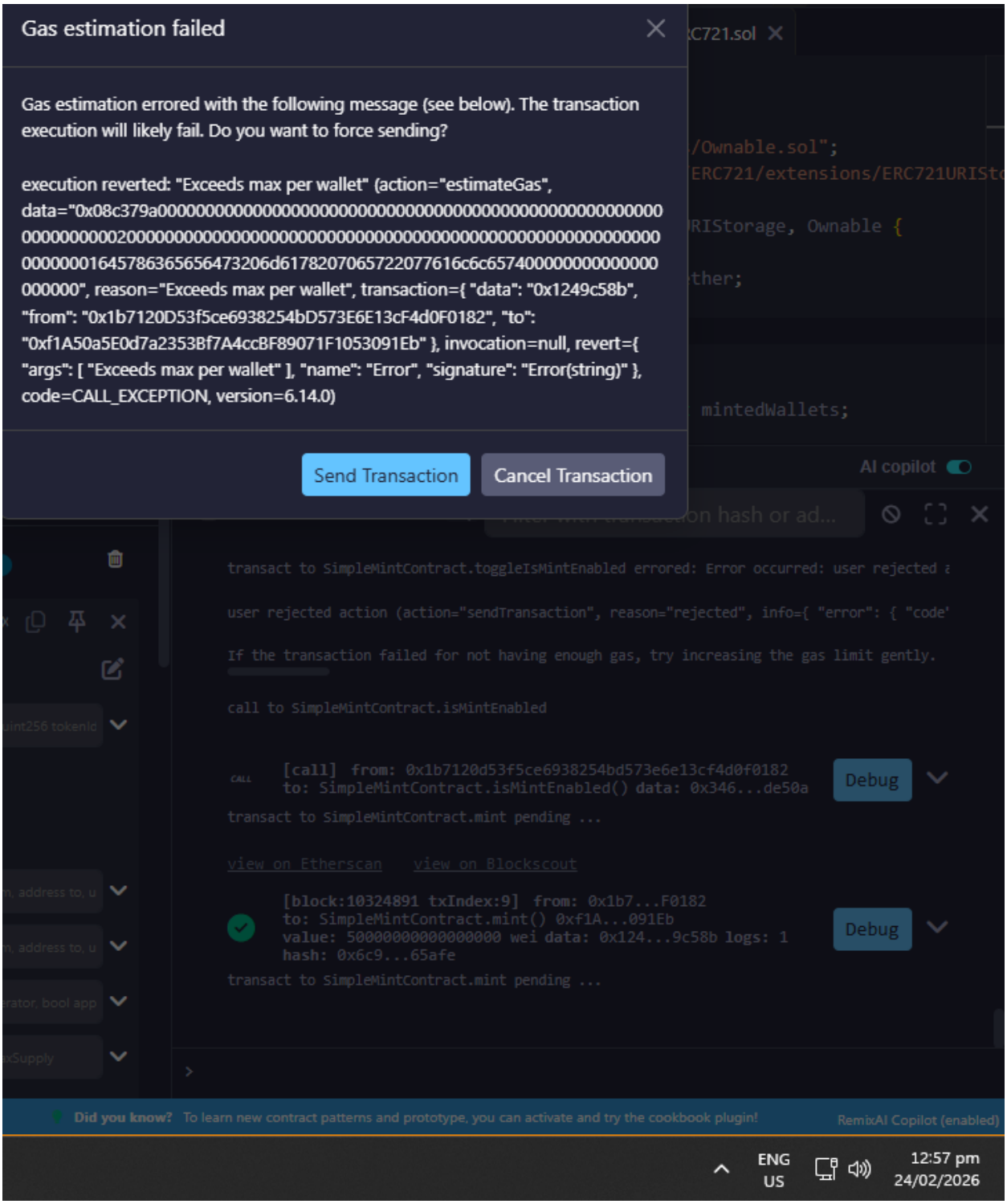




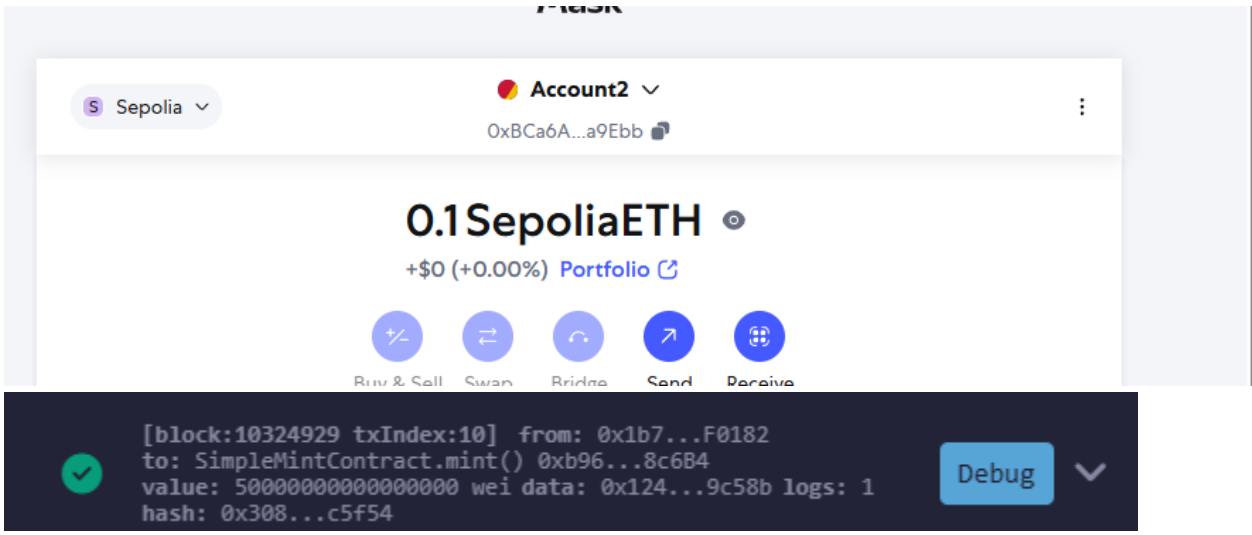
MINT SUCCESSFUL:

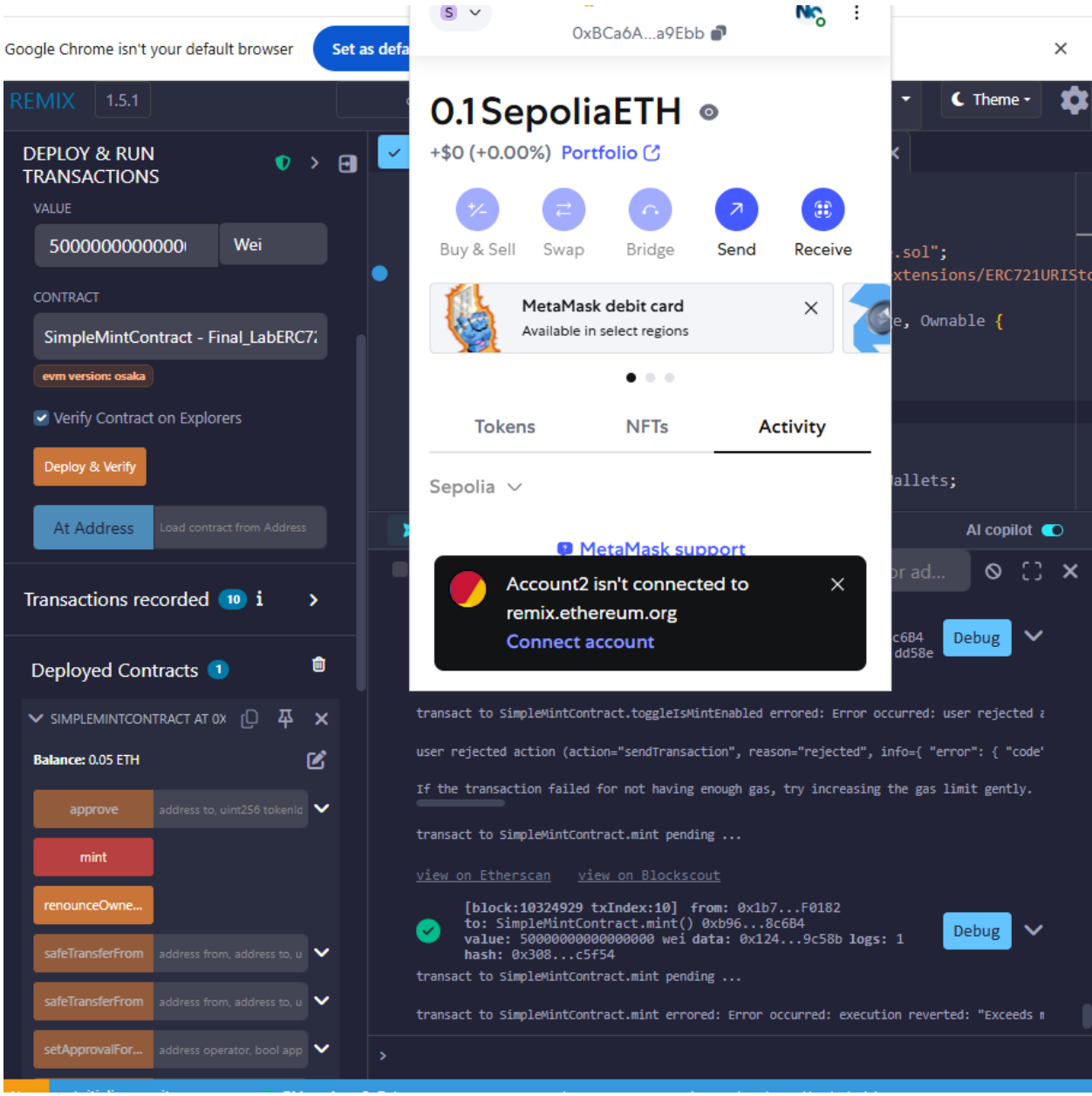

Tinkercad
Set as defa
Account 1
0x1b712...F0182
0.3637 SepoliaETH
+$0 (+0.00%) Portfolio
Buy & Sell
Swap
Bridge
Send
Receive
Ready to bridge?
Move across 9 chains, all within your wallet
Tokens
NFTs
Activity
Sepolia
Feb 24, 2026
Mint
Confirmed
-0.05 Sepolia...
-0.05 SepoliaETH
Toggle Is Mint Enabled
-0 SepoliaETH
Theme
AI copilot
If the transaction failed for not having enough gas, try increasing the gas limit gently.
call to SimpleMintContract.isMintEnabled
[call] from: 0x1b7120d53f5ce6938254bd573e6e13cf4d0f0182
to: SimpleMintContract.isMintEnabled() data: 0x346...de50a
Debug
transact to SimpleMintContract.mint pending ...
view on Etherscan
view on Blockscout
[block:10324891 txIndex:9] from: 0x1b7...F0182
to: SimpleMintContract.mint() 0xf1A...091Eb
value: 50000000000000000 wei data: 0x124...9c58b logs: 1
hash: 0x6c9...65afe
Debug

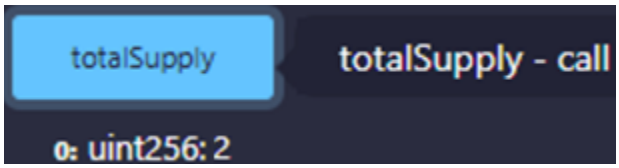
5. Revert: “Exceeds max per wallet”



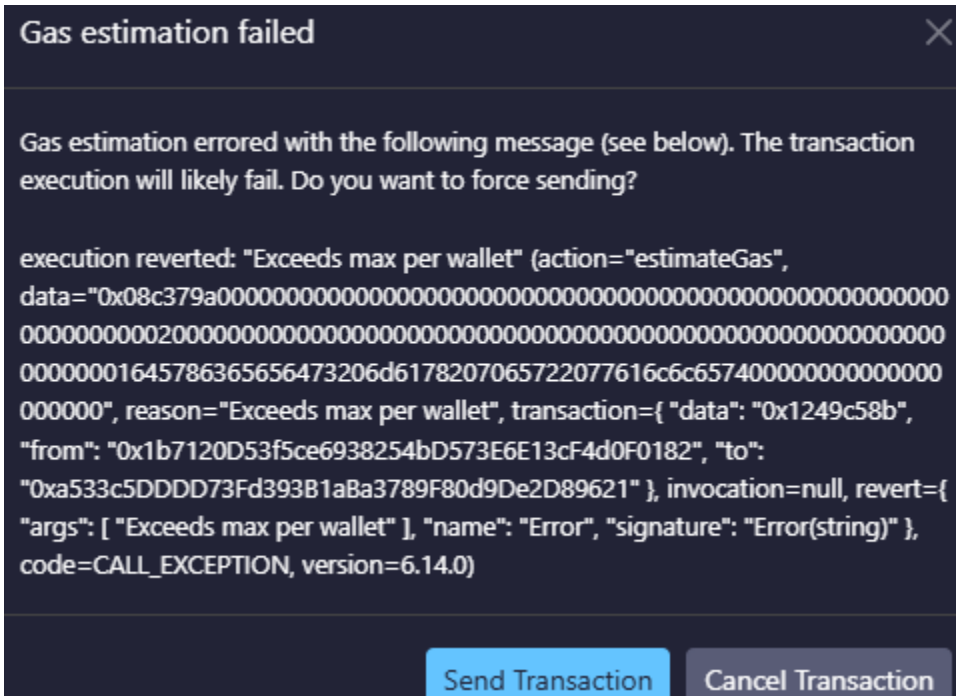
6. Successful mint transaction (2nd wallet)




Google Chrome isn't your default browser
Set as defa
0xBCa6A...a9Ebb
0.1SepoliaETH
+$0 (+0.00%) Portfolio
Buy & Sell
Swap
Bridge
Send
Receive
MetaMask debit card
Available in select regions
Tokens
NFTs
Activity
Sepolia
MetaMask support
Account2 isn't connected to remix.ethereum.org
Connect account
REMIX
1.5.1
Theme
DEPLOY & RUN TRANSACTIONS
VALUE
5000000000000
Wei
CONTRACT
SimpleMintContract - Final_LabERC7
evm version: osaka
Verify Contract on Explorers
Deploy & Verify
At Address
Load contract from Address
Transactions recorded 10
Deployed Contracts 1
SIMPLEMINTCONTRACT AT 0X
Balance: 0.05 ETH
approve
address to, uint256 tokenId
mint
renounceOwne...
safeTransferFrom
address from, address to, u
safeTransferFrom
address from, address to, u
setApprovalFor...
address operator, bool app
AI copilot
Debug
transact to SimpleMintContract.toggleIsMintEnabled errored: Error occurred: user rejected a
user rejected action (action="sendTransaction", reason="rejected", info={ "error": { "code'
If the transaction failed for not having enough gas, try increasing the gas limit gently.
transact to SimpleMintContract.mint pending ...
view on Etherscan
view on Blockscout
[block:10324929 txIndex:10] from: 0x1b7...F0182
to: SimpleMintContract.mint() 0xb96...8c6B4
value: 50000000000000000 wei data: 0x124...9c58b logs: 1
hash: 0x308...c5f54
Debug
transact to SimpleMintContract.mint pending ...
transact to SimpleMintContract.mint errored: Error occurred: execution reverted: "Exceeds m


totalSupply
totalSupply - call
0: uint256: 2

7. Revert: “Sold out”


Gas estimation failed
Gas estimation errored with the following message (see below). The transaction execution will likely fail. Do you want to force sending?
execution reverted: "Exceeds max per wallet" (action="estimateGas", data="0x08c379a0000000000000000000000000000000000000000000000000000000000000000000000000200000000000000000000000000000000000000000000000000000000000